Deep Reinforcement Learning

RL + DL

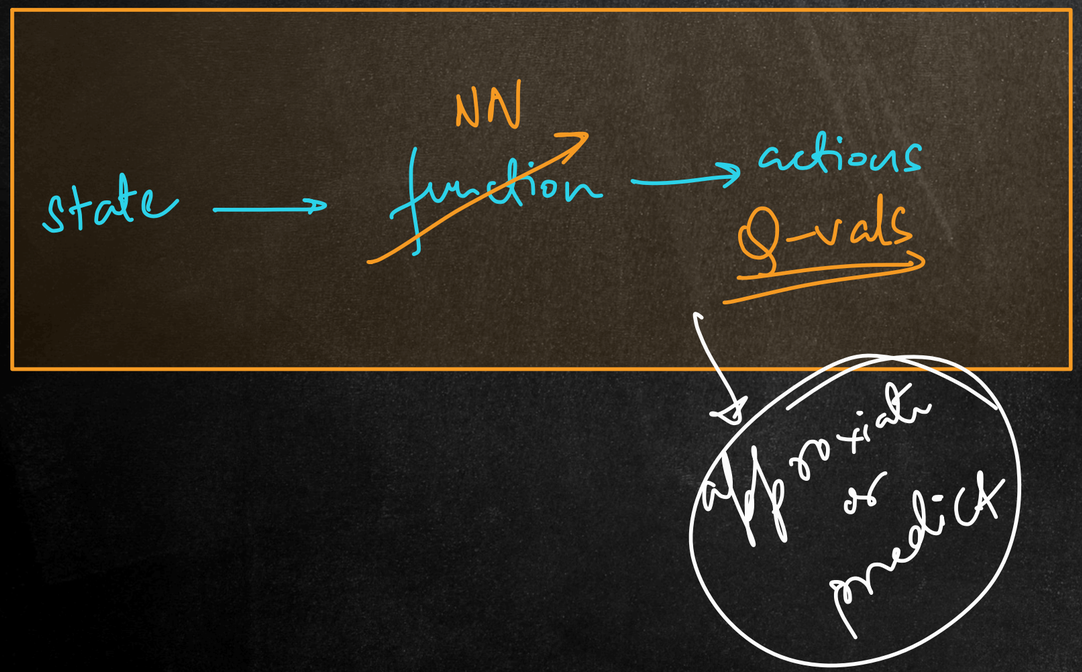
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Q-learning → Q-table Q(s,a) ⟹ 48 states
(discrete, small)

↓

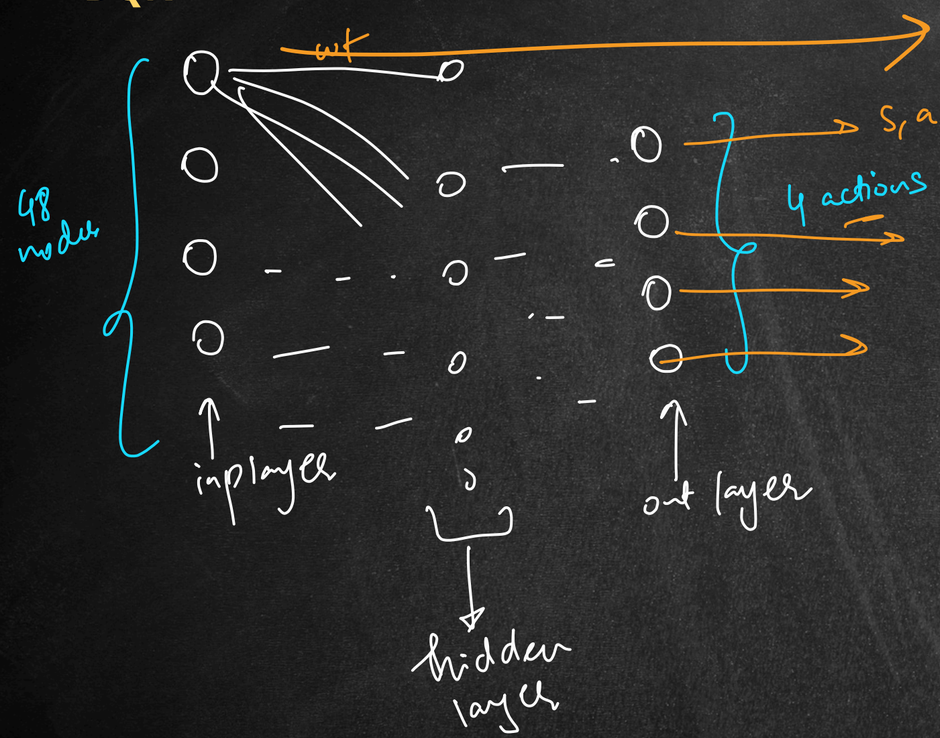
Deep Q-Networks (DQN)

problem
1. millions of states → huge
2. continuous state val → impossible



Deep Q-Networks

DQN



FP
1085
BP

48 × 4 ⟶ q-table
states action
q-val

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Q-learning ⇒ Q(s, a)

DQN ⇒ Q(s, a; θ)

NN parameters

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1. Correlated samples
2. Moving Target

Deep Q-Networks

Correlated Samples & Experience Replay

Naive DQN

experiences = State, Action, Reward, Next State, Terminal

Problem

NN → independent

① sequential
② correlated

batch size = 16

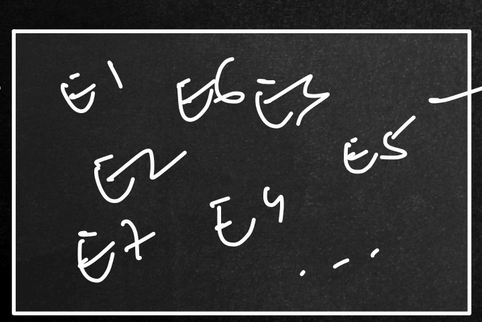
E1: S1 → A1 → R2 → S2
E2: S2 → A2 → R3 → S3
E3: S3 → A3 → R4 → S4

E1 → E2 → E3 ---

✗ E1 → E2 → E3 → E4 ----

✓ E9 → E121 → E2 → E5

Soln: ER ⇒ DQN memory (replay buffer)



→ randomly sample minibatch of Ex

Deep Q-Networks

Moving Target & Target Network

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Q-learning update $\Rightarrow$ $Q(s,a) = Q(s,a) + \alpha\left[r + \gamma \cdot \max_{a} Q(s',a) - Q(s,a)\right]$

leads to $\rightarrow$ unstable training

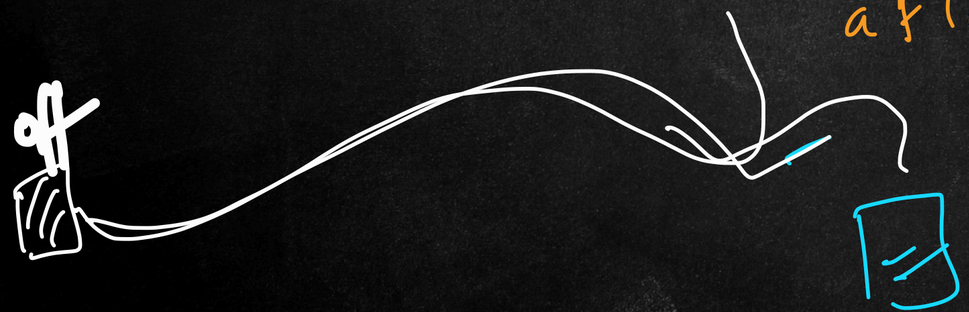
target $\Rightarrow$ $y = r + \gamma \cdot \max_{a} Q(s', a; \theta)$ $\rightarrow y$

predict Q-value $= Q(s', a; \theta) \rightarrow \hat{y}$

$loss = (y - \hat{y})$

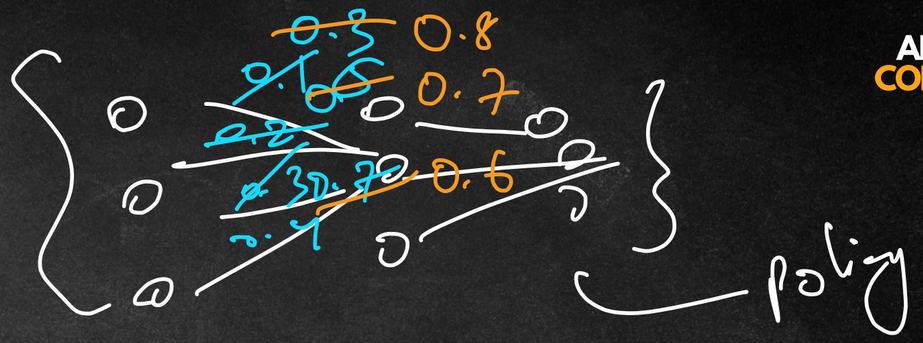
$=$

target is not a FIXED value



Deep Q-Networks

Moving Target & Target Network

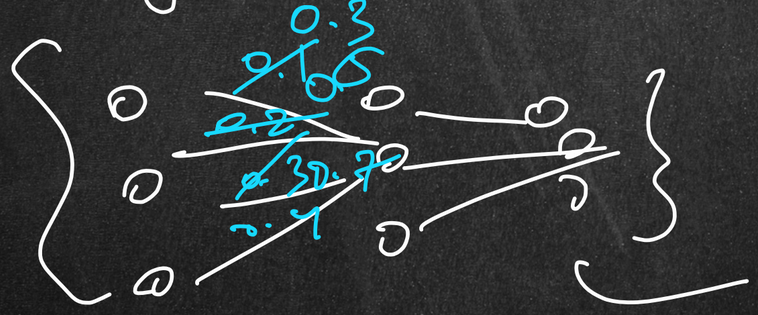


Naive DQN ⟶ 1 NN

DQN

① Policy Network (online)

② Target Network (old copy of policy network)



Architecture

single

Deep Q-Networks

Training Flappy Bird

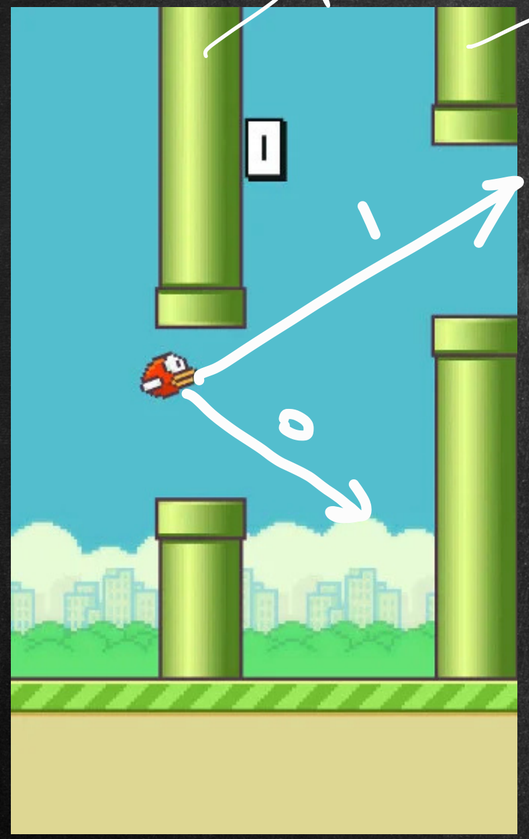
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Deep Q-Networks

Training Flappy Bird

Agent

Reinforcement learning
Minor Project

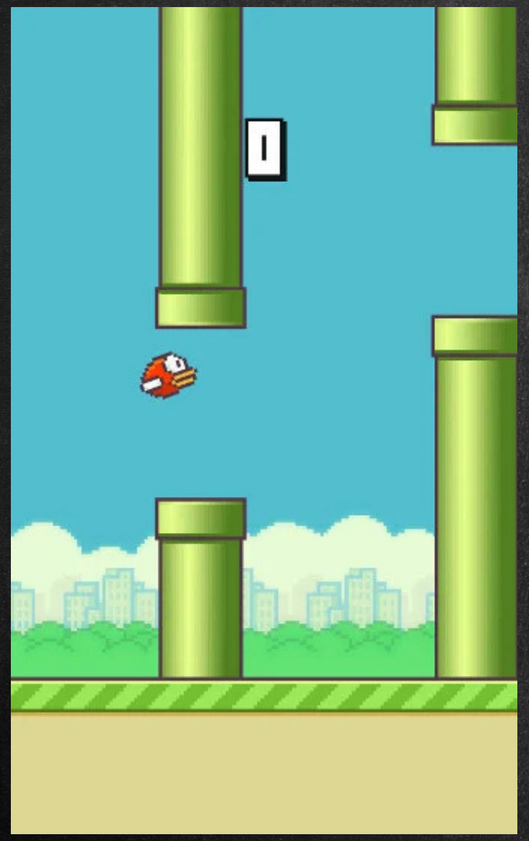


0 → do nothing
1 → flap

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Deep Q-Networks

Training Flappy Bird



LIDAR

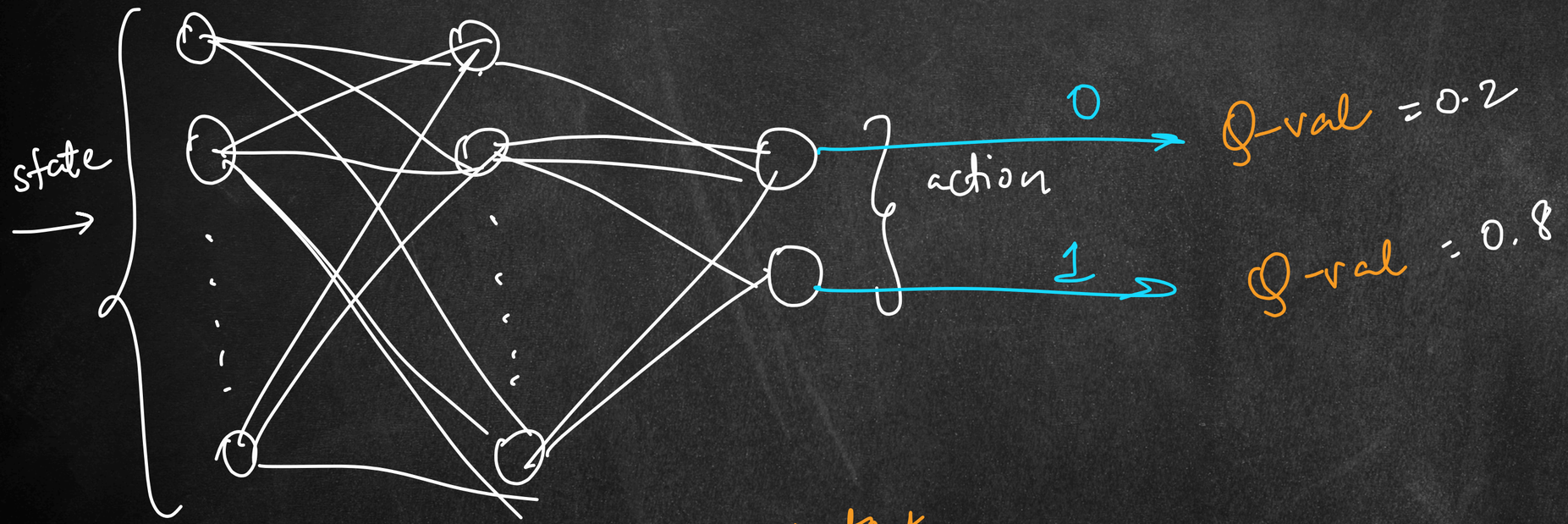
$\epsilon = 1$
↓
0.9
↓
0.8
↓
0.7
⋮
↓
0.05

0. Set Env
1. Define DQN
2. Create the Experience Replay
3. multiple episodes DQN train → setup
4. Epsilon-greedy policy + hyperparams (epsilon decay)
~~5~~. Target Network
8. Train & Test

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Deep Q-Networks

Training Flappy Bird → DQN → neural network arch



Inp
↓
12 nodes

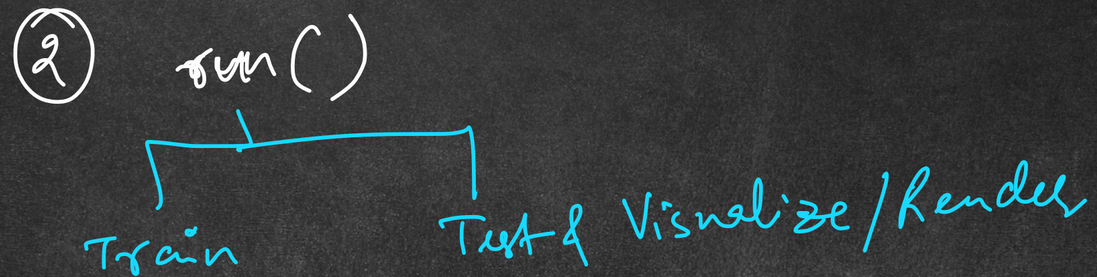
Hidden
↓
1 hidden layer
256 nodes

output
↓
2 nodes

Deep Q-Networks

Training Flappy Bird





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Deep Q-Networks

Training Flappy Bird

Experience Replay

Deque (double ended queue)



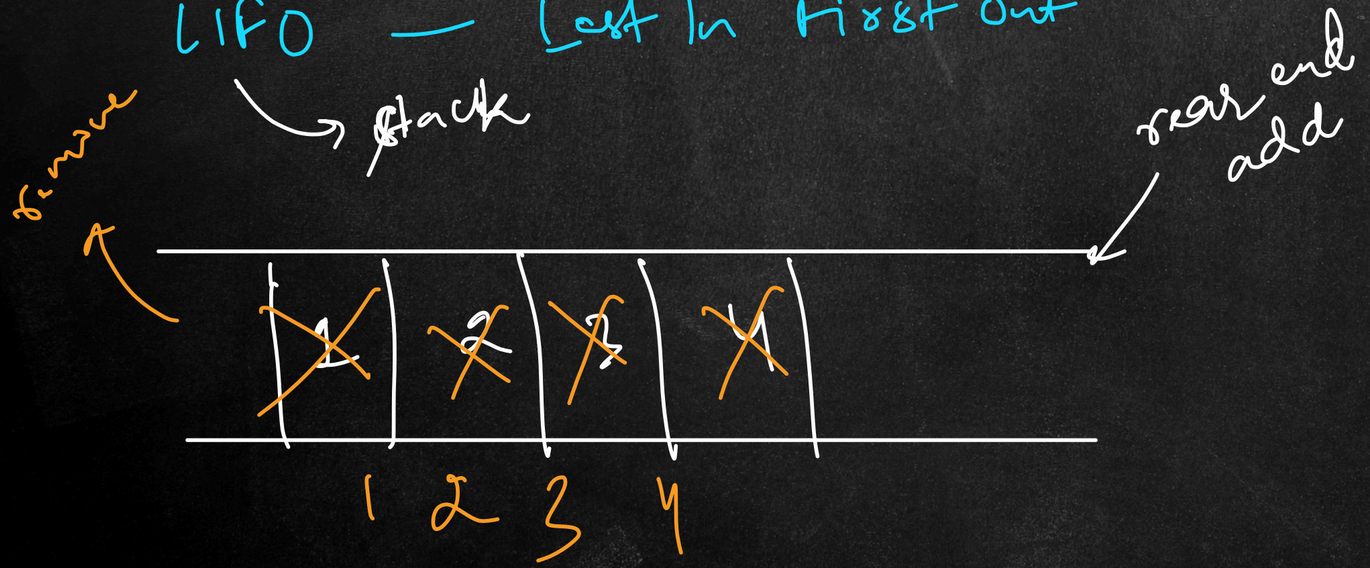
add: 1, 2, 3, 4

remove

FIFO mode — First In First Out

LIFO — Last In First Out

stack

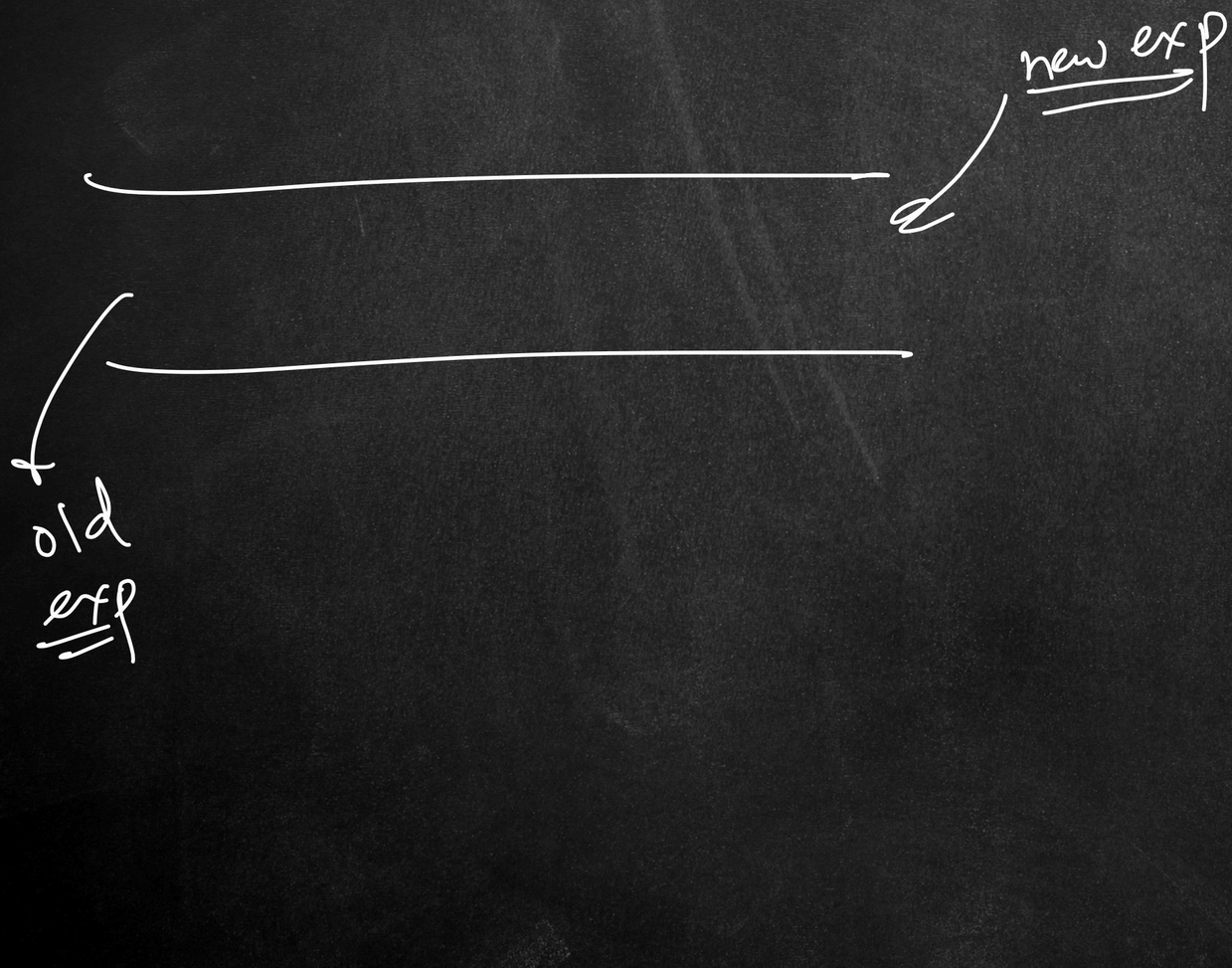


Deep Q-Networks

Training Flappy Bird

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experience : State, Action, Reward, Next State, Terminated



Deep Q-Networks

Training Flappy Bird

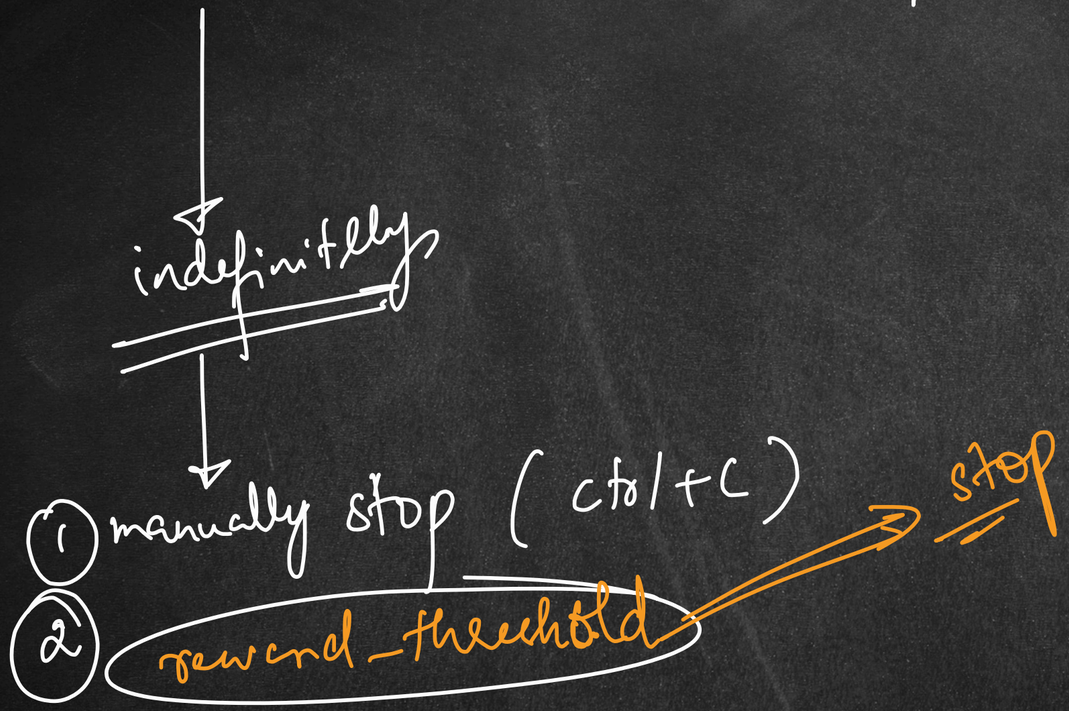
2000 ep

5000 ep



dqn 0





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Deep Q-Networks

Training Flappy Bird

→ ϵ- greedy policy

→ ϵ- decay

→ tensors

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